

Equations and Useful Information

$$N_A = 6.02 \times 10^{23} \text{ mole}^{-1} \quad k = 8.62 \times 10^{-5} \text{ eV/atom-K} \quad K = ^\circ\text{C} + 273$$

$$R = 1.987 \text{ cal/K-mole} = 8.314 \text{ J/K-mole} \quad 10^9 \text{ nm} = 10^6 \mu\text{m} = 10^2 \text{ cm} = 1 \text{ m}$$

$$1 \text{ GPa} = 10^3 \text{ MPa} = 10^9 \text{ Pa} = 10^9 \text{ N/m}^2 \quad 1 \text{ J/m}^2 = 1 \text{ N/m}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi R^3 \quad 1 \text{ Kg} = 1000 \text{ g} = 10^6 \text{ mg}$$

$$a_o = \frac{4R}{\sqrt{3}} \quad a_o = \frac{4R}{\sqrt{2}} = 2R\sqrt{2} \quad N_v = N e^{-Q_v / kT}$$

$$[uvw] \quad \langle uvw \rangle \quad (hkl) \quad \{hkl\} \quad \text{electron charge} = 1.6 \times 10^{-19} \text{ C}$$

$$N = \frac{N_A \rho}{A}$$

$$D = D_o e^{-Q_d/kT}$$

$$\nu = \frac{-\varepsilon_x}{\varepsilon_z} = \frac{-\varepsilon_y}{\varepsilon_z}$$

$$\sigma = E \varepsilon \quad U_r = \frac{\sigma_y \varepsilon_y}{2} = \frac{\sigma_y^2}{2E} \quad \sigma = \frac{F}{A_o} \quad \varepsilon = \frac{l - l_o}{l_o} = \frac{\Delta l}{l_o}$$

$$J = -D \frac{dC}{dx} \quad D = D_o \exp\left(\frac{-Q_d}{RT}\right) \quad \frac{C_x - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$E = 2G(1 + \nu) \quad TS = 3.45 \times HB \quad \sigma_c = \left(\frac{2E\gamma}{\pi a}\right)^{1/2} \quad \%CW = \frac{A_o - A_d}{A_o} \times 100$$

$$\%EL = \frac{l_f - l_o}{l_f} \times 100 \quad \%RA = \frac{A_o - A_f}{A_o} \times 100 \quad \sigma_y = \sigma_o + k_y d^{-1/2}$$

$$\sigma_m = \sigma_o \left(\frac{a}{\rho_t}\right)^{1/2} \quad K_{Ic} = Y\sigma\sqrt{\pi a}$$

$$\dot{\varepsilon}_s = K_2 \sigma^n \exp\left(\frac{-Q_d}{RT}\right) \quad \rho = \frac{n'(\sum A_c + \sum A_A)}{V_C N_A}$$

$$\text{Rate} = 1/\text{time} = A \exp(-Q/kT)$$

Atomic weights (g/mole): H = 1.0 C = 12.0 Cl = 35.5

$$V = IR \quad \rho = \frac{1}{\sigma} \quad \rho = \frac{RA}{l} \quad \sigma = n|e|\mu_e \quad \sigma = n|e|\mu_e + p|e|\mu_h$$

$$V_m + V_f = 1 \quad E_c(l) = \frac{E_m E_p}{E_p V_m + E_m V_p} \quad E_{cl} = E_m V_m + E_f V_f$$

$$l_c = \frac{\sigma_f^* d}{2\tau_c} \quad \frac{1}{E_{ct}} = \frac{V_m}{E_m} + \frac{V_f}{E_f} \quad \frac{F_f}{F_m} = \frac{E_f V_f}{E_m V_m} \quad \sigma_{cl} = \sigma'_m(1 - V_f) + \sigma_f^* V_f$$

ERROR FUNCTION TABLE

z	erf(z)	z	erf(z)	z	erf(z)	z	erf(z)
0.20	0.2227	0.55	0.5633	0.90	0.7970	1.50	0.9661
0.25	0.2763	0.60	0.6039	0.95	0.8209	1.60	0.9763
0.30	0.3286	0.65	0.6420	1.00	0.8427	1.70	0.9838
0.35	0.3794	0.70	0.6778	1.10	0.8802	1.80	0.9891
0.40	0.4284	0.75	0.7712	1.20	0.9103	1.90	0.9928
0.45	0.4755	0.80	0.7421	1.30	0.9340	2.00	0.9953
0.50	0.5205	0.85	0.7707	1.40	0.9523	2.20	0.9981

